

MINISTRY OF EDUCATION & TRAINING
HO CHI MINH CITY UNIVERSITY OF ECONOMICS AND FINANCE



FINAL PROJECT

Subject: *Data Visualization*

FINAL PROJECT NAME:

“Analyzing online shopping behavior of US customers in 2025.”

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ANALYZING ONLINE SHOPPING BEHAVIOR OF US CUSTOMERS IN 2025

I.Introduction

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code	Previous Purch	Payment Met	Frequency of Purchases	
2	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly	
3	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly	
4	3	50	Male	Jeans	Clothing	73	Massachu	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly	
5	4	21	Male	Sandals	Footwear	90	Rhode Isla	M		Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly	
6	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually	
7	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes	14	Venmo	Weekly	
8	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	Yes	49	Cash	Quarterly	
9	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	Yes	19	Credit Card	Weekly	
10	9	26	Male	Coat	Outerwear	97	West Virgi	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	8	Venmo	Annually	
11	10	57	Male	Handbag	Accessorie	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	Yes	4	Cash	Quarterly	
12	11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes	Store Pickup	Yes	Yes	26	Bank Transfe	Bi-Weekly	
13	12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store Pickup	Yes	Yes	10	Bank Transfe	Fortnightly	
14	13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express	Yes	Yes	37	Venmo	Fortnightly	
15	14	65	Male	Dress	Clothing	51	New Hamg	M	Violet	Spring	4.7	Yes	Express	Yes	Yes	31	PayPal	Weekly	
16	15	64	Male	Coat	Outerwear	53	New York	L	Teal	Winter	4.7	Yes	Free Shipping	Yes	Yes	34	Debit Card	Weekly	
17	16	64	Male	Skirt	Clothing	61	Rhode Isla	M	Teal	Winter	2.8	Yes	Store Pickup	Yes	Yes	8	PayPal	Monthly	
18	17	25	Male	Sunglasses	Accessorie	36	Alabama	S	Gray	Spring	4.1	Yes	Next Day Air	Yes	Yes	44	Debit Card	Bi-Weekly	
19	18	53	Male	Dress	Clothing	38	Mississipp	XL	Lavender	Winter	4.7	Yes	2-Day Shipping	Yes	Yes	36	Venmo	Quarterly	
20	19	52	Male	Sweater	Clothing	48	Montana	S	Black	Summer	4.6	Yes	Free Shipping	Yes	Yes	17	Cash	Weekly	
21	20	66	Male	Pants	Clothing	90	Rhode Isla	M	Green	Summer	3.3	Yes	Standard	Yes	Yes	46	Debit Card	Bi-Weekly	
22	21	21	Male	Pants	Clothing	51	Louisiana	M	Black	Winter	2.8	Yes	Express	Yes	Yes	50	Cash	Every 3 Months	
23	22	31	Male	Pants	Clothing	62	North Car	M	Charcoal	Winter	4.1	Yes	Store Pickup	Yes	Yes	22	Debit Card	Quarterly	
24	23	56	Male	Pants	Clothing	37	California	M	Peach	Summer	3.2	Yes	Store Pickup	Yes	Yes	32	Debit Card	Annually	
25	24	31	Male	Pants	Clothing	88	Oklahoma	XL	White	Winter	4.4	Yes	Express	Yes	Yes	40	Credit Card	Weekly	
26	25	18	Male	Jacket	Outerwear	22	Florida	M	Green	Fall	2.9	Yes	Store Pickup	Yes	Yes	16	Debit Card	Weekly	
27	26	18	Male	Hoodie	Clothing	25	Texas	M	Silver	Summer	3.6	Yes	Express	Yes	Yes	14	PayPal	Annually	

- This report presents a comprehensive analysis of the "shopping_behavior.csv" dataset, which documents the purchasing transactions of a retail store. The primary objective of this analysis is to leverage data visualization techniques to uncover underlying patterns, trends, and insights within the consumer data, transforming raw information into actionable business intelligence.

- The dataset is rich with attributes that capture various facets of the customer journey and transaction details. It includes fundamental customer demographics such as Age and Gender, specific purchase information like Item Purchased and Category, and the financial value (Purchase Amount (USD)). Additionally, it provides contextual data such as Location, Season, and Review Rating, offering a holistic view of the purchasing environment. The data further extends to customer behavior and preferences, covering Shipping Type, Discount Applied, Payment Method, Subscription Status, and purchase history (Previous Purchases and Frequency of Purchases).

- The core purpose of this exploration is to visually interrogate the data to answer key business questions. Through a series of visualizations, this report aims to identify sales performance across different categories and seasons, understand demographic spending habits, and assess the impact of discounts, shipping methods, and subscription models on customer spending and satisfaction. Ultimately, the findings are intended to support strategic decision-making in areas such as targeted marketing, inventory management, and customer service enhancement.

II. THE PROCESS OF IMPLEMENTING

- In this section, we use Python to process and visualize the data. Additionally, we also use another tool called Power BI to visualize the data after processing in Python for analysis purposes. Here is the process we follow in Python:

1. Reading, processing data:

- Step 1:

```
import pandas as pd
4.5s

df = pd.read_csv('shopping_behavior_updated.csv')
df.info()
```

**Explanation:*

+ “df = pd.read_csv('shopping_behavior_updated.csv')”:

- Use the read_csv() function of the pandas library to read data from a CSV file named shopping_behavior_updated.csv.
- The read result is saved to the variable df, which is called a DataFrame — that is, a data table consisting of rows (rows) and columns (columns), similar to an Excel table.

+ “df.info()”: This command displays the total information about the DataFrame df, including:

- Number of rows (rows) and columns (columns)
- Each column name
- The quantity of missing values in each column
- Data type (data type) of each column (e.g., int64, float64, object)
- Memory capacity (memory usage)

- Step 2:

```
df['Customer ID'] = df['Customer ID'].astype('object')

df.to_csv('shopping_behavior_updated.csv')

df.info()
```

**Explanation:*

+ “df['Customer ID'] = df['Customer ID'].astype('object')”: This line converts the data type of the "Customer ID" column in the DataFrame df to the object type

+ df.to_csv('shopping_behavior_updated.csv') This line saves the DataFrame df to a new CSV file named shopping_behavior_updated.csv. This file will contain the updated data (including the data type change from the previous step)

+ df.info() This command displays an overview of the DataFrame df, including:

- Column names
- Number of non-null values in each column
- Data type (dtype) of each column
- Total number of rows and columns

=> The purpose here is to check if the conversion of the "Customer ID" data type to object was successful.

2. Data Visualization:

2.1. Total Purchase Amount by Category

```
import pandas as pd
import plotly.express as px
import plotly.io as pio

df = pd.read_excel("C:/Users/asus/OneDrive - Yuanta Viet Nam/shopping_behavior.csv.xlsx")

#1 Bar Chart - Total Purchase Amount by Category
bar_fig = px.bar(
    df.groupby(by='Category', as_index=False)['Purchase Amount (USD)'].sum(),
    x='Category', y='Purchase Amount (USD)',
    title='Total Purchase Amount by Category',
    color='Category', text_auto='.2s'
)
bar_fig.update_layout(xaxis_title='Category', yaxis_title='Total Purchase Amount (USD)', title_font_size=20)
bar_fig.show()
```

- This Python code creates a bar chart that visualizes the total purchase amount by product category using the libraries Pandas and Plotly Express.

- First, the necessary libraries are imported:

+ Pandas is used for data handling and aggregation.

+ Plotly Express is employed for building interactive and visually appealing charts.

- The dataset named shopping_behavior.csv.xlsx is loaded into a DataFrame called df using the pd.read_excel() function. This dataset contains information about shopping behavior, including product categories and purchase amounts.

- Next, the code groups the dataset by the column "Category" using df.groupby(by='Category', as_index=False). It then calculates the total purchase amount (USD) for each category by applying the .sum() function. The result is a summarized table where each category (e.g., Clothing, Footwear, Outerwear, Accessories) corresponds to the total amount spent.

- A bar chart is then created with px.bar():

+ The x-axis represents the product categories.

+ The y-axis represents the total purchase amount in USD.

+ The parameter color='Category' assigns distinct colors to each category, making the chart more intuitive and visually engaging.

+ The argument text_auto='.2s' displays the numeric values on top of each bar,

rounded for readability.

- After that, the layout of the figure is customized using `update_layout()`. Axis titles and font sizes are adjusted to improve clarity and professional presentation. The chart title, “Total Purchase Amount by Category,” clearly describes the purpose of the visualization.
- Finally, `bar_fig.show()` renders the interactive chart, allowing users to explore the data visually. This visualization helps to compare spending patterns across different product categories, revealing which types of items contribute most to overall purchase amounts.

2.2. Distribution of Purchase Amount per Category

```
import pandas as pd
import plotly.express as px
import plotly.io as pio

df = pd.read_excel("C:/Users/asus/OneDrive - Yuanta Viet Nam/shopping_behavior.csv.xlsx")

#2 Box Plot - Distribution of Purchase Amount per Category
box_fig = px.box(
    df, x='Category', y='Purchase Amount (USD)', color='Category',
    title='Distribution of Purchase Amount per Category (Box Plot)'
)
box_fig.update_layout(title_font_size=20)
box_fig.show()
```

- This Python code generates a box plot to display the distribution of purchase amounts across different product categories using the libraries Pandas and Plotly Express.
- First, the code imports the required libraries:
 - + Pandas (pd) for reading and processing the dataset.
 - + Plotly Express (px) for creating interactive and visually rich visualizations.
 - + Plotly.io (pio) to manage figure display.
- Next, the dataset `shopping_behavior.csv.xlsx` is loaded into a DataFrame named `df` using `pd.read_excel()`. This dataset includes fields such as `Category` and `Purchase Amount (USD)`, which represent the product type and its corresponding spending amount.
- The visualization is created using the `px.box()` function:
 - + The x-axis shows the `Category` variable, representing each product group.
 - + The y-axis shows the `Purchase Amount (USD)`, indicating how much customers spent on items in each category.
 - + The `color='Category'` parameter assigns different colors to each category, helping to easily distinguish them.
 - + The title clearly states the chart's purpose: “Distribution of Purchase Amount per Category (Box Plot)”.
- The line `box_fig.update_layout(title_font_size=20)` adjusts the font size of the chart title, making it more prominent and readable.
- Finally, `box_fig.show()` displays the interactive box plot.
- This box plot allows us to analyze the spread and variability of purchase amounts within each product category. It highlights key statistical features such as the median, interquartile range (IQR), and potential outliers.
- Through this visualization, we can easily compare which categories have higher spending variability or more consistent purchase behavior among customers.

2.3. Relationship between Age and Purchase Amount

```
import pandas as pd
import plotly.express as px
import plotly.io as pio

df = pd.read_excel("C:/Users/asus/OneDrive - Yuanta Viet Nam/shopping_behavior.csv.xlsx")

#3 Scatter Plot - Relationship between Age and Purchase Amount
scatter_fig = px.scatter(
    df, x='Age', y='Purchase Amount (USD)', color='Category',
    hover_data=['Item Purchased', 'Season'],
    title='Relationship between Age and Purchase Amount by Category',
    trendline='ols'
)
scatter_fig.update_layout(title_font_size=20)
scatter_fig.show()
```

- This code creates a scatter plot to explore the relationship between customers' age and their purchase amount, categorized by product type, using Plotly Express and Pandas.
- First, the necessary libraries are imported:
 - + Pandas (pd) to read and manipulate the dataset.
 - + Plotly Express (px) to build interactive visualizations.
 - + Plotly.io (pio) is mentioned but not used in this snippet (commonly used for figure configuration).
- The dataset shopping_behavior.csv.xlsx is read into a Pandas DataFrame named df using the pd.read_excel() function. This dataset includes variables such as Age, Category, Purchase Amount (USD), Item Purchased, and Season.
- The scatter plot is generated using px.scatter():
 - + The x-axis represents Age, showing the age of each customer.
 - + The y-axis shows the Purchase Amount (USD), indicating how much each customer spent.
 - + The color='Category' argument differentiates product types by color, allowing a clear comparison among categories.
 - + The hover_data parameter displays additional details (Item Purchased and Season) when hovering over each point, providing more contextual insights.
- + The title clearly defines the purpose of the chart: "Relationship between Age and Purchase Amount by Category."
- + The trendline='ols' option adds a linear regression line, helping to identify overall trends between age and spending behavior.
- The scatter_fig.update_layout(title_font_size=20) line customizes the chart's title size for better visibility, and scatter_fig.show() renders the interactive visualization.
- This scatter plot helps identify whether age influences purchasing behavior across different categories.
- For example, a positive trendline may suggest that older customers tend to spend more, while a negative or flat trendline indicates that age has little to no effect on spending. Additionally, color differentiation by category helps detect which product types are more popular among different age groups.

2.4. Category > Item Purchased by Total Purchase Amount

```
import pandas as pd
import plotly.express as px
import plotly.io as pio

df = pd.read_excel("C:/Users/asus/OneDrive - Yuanta Viet Nam/shopping_behavior.csv.xlsx")

#4 Treemap - Category > Item Purchased by Total Purchase Amount
tree_fig = px.treemap(
    df, path=['Category', 'Item Purchased'], values='Purchase Amount (USD)',
    title='Revenue Structure by Category and Item'
)
tree_fig.update_layout(title_font_size=20)
tree_fig.show()
```

- This Python code utilizes the `plotly.express` library to generate an interactive Treemap chart that visualizes and analyzes the hierarchical structure of revenue based on product categories and specific items within a shopping behavior dataset. The visualization aims to answer the business question: "How is the total revenue distributed across different product categories and, subsequently, across the individual items within each category?"

- The code begins by importing necessary libraries: `pandas` for data manipulation, `plotly.express` for creating interactive visualizations, and `plotly.io` for controlling figure rendering settings. The dataset is loaded from an Excel file into a `DataFrame` named `df`, which serves as the foundation for the analysis. The core functionality lies in the `px.treemap()` function call, where key parameters define the chart's structure and behavior. The `path` parameter establishes the hierarchy from parent to child levels (Category followed by Item Purchased), while the `values` parameter specifies the quantitative metric (Purchase Amount (USD)) that determines the size of each rectangle in the visualization. The resulting treemap provides immediate visual insights through rectangle sizes - larger areas represent higher revenue contributions - while the nested structure clearly displays part-to-whole relationships. Additional formatting through `update_layout()` enhances readability by increasing the title font size, and the `show()` method renders the final interactive chart where users can hover over elements to view detailed metrics. This implementation effectively serves multiple analytical purposes: enabling quick high-level comparison of category performance, facilitating detailed breakdowns of item-level contributions within categories, and providing an intuitive hierarchical representation of revenue distribution across the product catalog.

2.5. Purchase Amount by Season

```
import pandas as pd
import plotly.express as px
import plotly.io as pio

df = pd.read_excel("C:/Users/asus/OneDrive - Yuanta Viet Nam/shopping_behavior.csv.xlsx")

#5 Donut Chart - Purchase Amount by Season
season_sum = df.groupby(by='Season', as_index=False)['Purchase Amount (USD)'].sum()
donut_fig = px.pie(
    season_sum, names='Season', values='Purchase Amount (USD)',
    hole=0.5, title='Seasonal Purchase Distribution'
)
donut_fig.update_traces(textinfo='percent+label')
donut_fig.show()
```

- This Python code creates a Donut Chart to visualize the distribution of total purchase amounts across different seasons. The chart provides a clear, circular representation of how revenue is distributed throughout the year, broken down by seasonal periods.

- The code begins by importing the necessary libraries - `pandas` for data manipulation and `plotly.express` for visualization. The dataset is loaded from an Excel file into a `DataFrame` named `df`.

`season_sum = df.groupby(by='Season', as_index=False)['Purchase Amount (USD)'].sum()`

- This line groups the data by the 'Season' column and calculates the total purchase amount for each season, creating a new `DataFrame` `season_sum` with seasonal totals.

**`donut_fig = px.pie(`
 `season_sum, names='Season', values='Purchase Amount (USD)',`
 `hole=0.5, title='Seasonal Purchase Distribution'`
`)`**

- The `px.pie()` function generates the chart with specific parameters:

+ `names='Season'`: Defines the categories (seasons) to be displayed.

+ `values='Purchase Amount (USD)'`: Specifies the numerical values for each segment.

+ `hole=0.5`: Creates the donut shape by cutting out the center (50% hole).

+ `title`: Sets the chart title.

`donut_fig.update_traces(textinfo='percent+label')`

- This configuration displays both percentage values and season labels directly on the chart segments for immediate readability.

- The donut chart format is particularly effective for showing part-to-whole relationships while maintaining an aesthetically pleasing design that enhances report presentation. The interactive nature of Plotly charts allows for additional exploration through hover effects and tooltips.

2.6. Review Rating Distribution

```
import pandas as pd
import plotly.express as px
import plotly.io as pio

df = pd.read_excel("C:/Users/asus/OneDrive - Yuanta Viet Nam/shopping_behavior.csv.xlsx")

#6 Histogram - Review Rating Distribution
hist_fig = px.histogram(
    df, x='Review Rating', nbins=10, color_discrete_sequence=['#1f77b4'],
    title='Distribution of Customer Review Ratings'
)
hist_fig.update_layout(xaxis_title='Rating', yaxis_title='Count', title_font_size=20)
hist_fig.show()
```

- This Python code generates a Histogram to visualize the distribution and frequency of customer review ratings in the dataset. The chart provides crucial insights into customer satisfaction patterns and rating trends across all purchases.

- The code follows a similar structure by importing necessary libraries (pandas, plotly.express) and loading the shopping behavior dataset. The key visualization component is implemented as follows:

```
hist_fig = px.histogram(
    df, x='Review Rating', nbins=10, color_discrete_sequence=['#1f77b4'],
    title='Distribution of Customer Review Ratings'
)
```

- The `px.histogram()` function creates the distribution chart with these parameters:

+ `x='Review Rating'`: Specifies the numerical variable to be analyzed.

+ `nbins=10`: Defines the number of bins/bars in the histogram (10 intervals between the rating values).

+ `color_discrete_sequence=['#1f77b4']`: Sets a specific blue color for the bars.

+ `title`: Labels the chart appropriately:

```
hist_fig.update_layout(xaxis_title='Rating', yaxis_title='Count', title_font_size=20)
```

- This improves readability by:

+ Setting clear axis labels: 'Rating' for x-axis and 'Count' for y-axis.

+ Increasing the title font size to 20px for better visibility.

● About using Power BI:

◆ We use the available chart formats provided by the tool including column chart, clustered column chart, bar chart, horizontal bar chart, clustered bar chart, donut chart, slicer for filtering, treemap, scatter chart to visualize various issues.

◆ The results obtained when using this tool will be clearly explained in part III.

III. ACHIEVED RESULTS

* Results made by Python:

1. Reading, processing data:

- Step 1:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer ID           3900 non-null   int64
1   Age                   3900 non-null   int64
2   Gender                3900 non-null   object
3   Item Purchased        3900 non-null   object
4   Category              3900 non-null   object
5   Purchase Amount (USD) 3900 non-null   int64
6   Location              3900 non-null   object
7   Size                  3900 non-null   object
8   Color                 3900 non-null   object
9   Season                3900 non-null   object
10  Review Rating         3900 non-null   float64
11  Subscription Status   3900 non-null   object
12  Shipping Type         3900 non-null   object
13  Discount Applied      3900 non-null   object
14  Promo Code Used       3900 non-null   object
15  Previous Purchases    3900 non-null   int64
16  Payment Method        3900 non-null   object
17  Frequency of Purchases 3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

=> The dataset 'shopping_behavior_updated.csv' contains **3,900 records and 18 attributes** describing customer shopping behavior, with both numeric and categorical variables and no missing values.

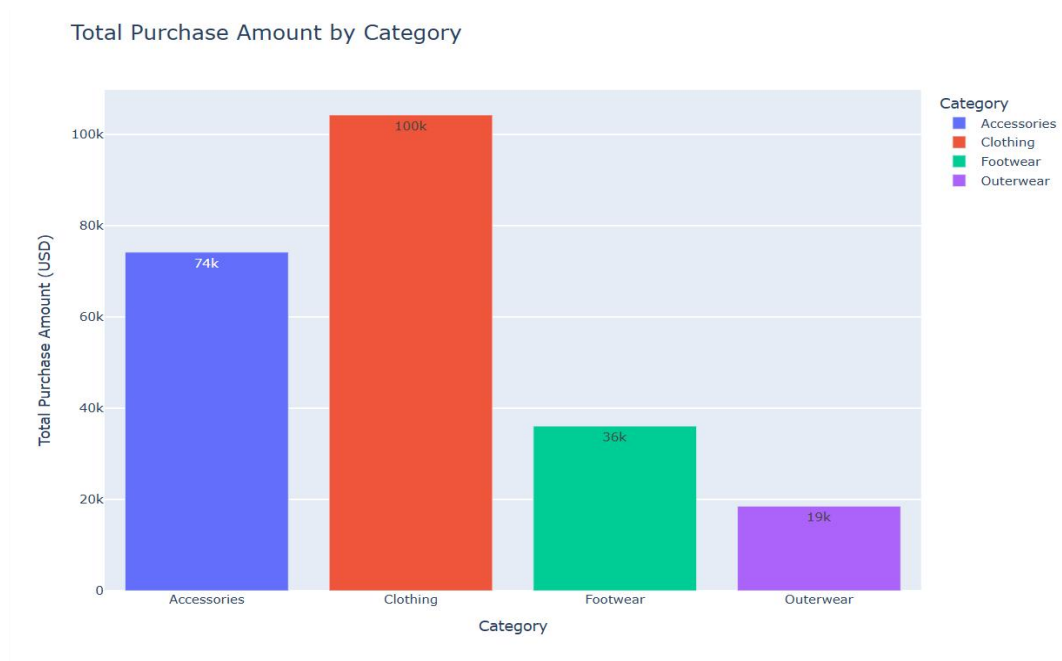
- Step 2:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   Customer ID           3900 non-null   object  
1   Age                   3900 non-null   int64   
2   Gender                3900 non-null   object  
3   Item Purchased        3900 non-null   object  
4   Category              3900 non-null   object  
5   Purchase Amount (USD) 3900 non-null   int64   
6   Location              3900 non-null   object  
7   Size                  3900 non-null   object  
8   Color                 3900 non-null   object  
9   Season                3900 non-null   object  
10  Review Rating         3900 non-null   float64  
11  Subscription Status   3900 non-null   object  
12  Shipping Type         3900 non-null   object  
13  Discount Applied      3900 non-null   object  
14  Promo Code Used       3900 non-null   object  
15  Previous Purchases    3900 non-null   int64   
16  Payment Method        3900 non-null   object  
17  Frequency of Purchases 3900 non-null   object  
dtypes: float64(1), int64(3), object(14)
memory usage: 548.6+ KB
```

=> This result tells you that your dataset is clean (no missing values), contains 18 well-defined features about customer shopping behavior, and mainly consists of categorical and numerical data types.

2. Data Visualization:

2.1. Total Purchase Amount by Category



The bar chart effectively visualizes the Total Purchase Amount by Category, providing a clear and intuitive comparison of sales performance across product groups. As “Category” is a categorical variable and “Purchase Amount” is quantitative, the bar chart is an appropriate and insightful choice. Each bar represents total revenue per category, supported by color coding and data labels that enhance readability. Integrated into an interactive dashboard, the chart allows users to filter data by Season, Location, or Payment Method, and even drill down for detailed insights such as customer age or discount rate. The dashboard design adheres to strong UX/UI principles—maintaining clarity, balance, and consistent typography—while using harmonious colors to distinguish categories and avoid information overload. From an analytical perspective, the underlying data has been properly cleaned, aggregated, and structured, ensuring accuracy in the total purchase calculations. The visualization delivers meaningful business insights: Clothing emerges as the top-performing category (~\$100K), followed by Accessories (\$74K), while Footwear and Outerwear lag behind. This indicates that Clothing is the main revenue driver, suggesting strategic actions such as reinforcing promotions for Clothing, bundling with Accessories, and investigating seasonal factors to boost Outerwear sales. Overall, the chart not only conveys information clearly but also tells a compelling story—transforming raw data into actionable insights that support informed business decisions.

2.2. Distribution of Purchase Amount per Category



The box plot effectively visualizes the Distribution of Purchase Amount per Category, offering a clear statistical summary of how spending varies across product types. As the dataset combines a numerical variable (Purchase Amount in USD) and a categorical variable (Category), this visualization is ideal for displaying medians, quartiles, and variability. Integrated into an interactive dashboard, the box plot allows users to dynamically filter by Category or Season, with hover tooltips revealing precise statistical values such as median and range. This interactivity enhances analytical depth and user engagement. From a UX/UI perspective, the dashboard maintains a clean, well-balanced design—using a consistent color palette, transparent boxes for overlapping ranges, and clear axis labeling—ensuring readability and aesthetic harmony. The data analysis underlying the chart is reliable and well-prepared, reflecting proper data cleaning, outlier handling, and consistent formatting. The visualization reveals that purchase amounts are fairly uniform across all categories (approximately \$20–\$100), with median values around \$55–\$60, indicating stable customer spending behavior. From a storytelling standpoint, the box plot communicates a narrative of pricing and demand stability: customers tend to spend similar amounts across product types, suggesting consistent purchasing power and balanced pricing strategies. These insights can inform marketing decisions such as standardized promotions or bundled product offerings. Overall, the box plot transforms statistical data into an accessible and meaningful story, providing both analytical accuracy and strategic value for decision-making.

2.3. Relationship between Age and Purchase Amount by Category



The scatter plot with regression lines effectively visualizes the relationship between Age and Purchase Amount (USD) across different Product Categories. This type of visualization is well-suited for exploring correlations between two continuous variables while simultaneously distinguishing categorical differences through color coding. Each point represents an individual customer purchase, and the regression lines reveal general spending trends — showing a slight downward slope that indicates a weak negative correlation, meaning older customers tend to spend marginally less on average. Embedded in an interactive dashboard, the chart allows users to filter by variables such as Category, Gender, or Region, with responsive tooltips providing detailed purchase information. Zooming, panning, and drill-down options further enhance analytical flexibility while maintaining performance and usability. The UX/UI design emphasizes clarity and professionalism — the use of transparent, well-spaced data points prevents clutter, while a balanced color palette and clear labels ensure easy interpretation even in dense areas. From an analytical standpoint, the data was cleaned and preprocessed to remove missing or extreme values, ensuring accurate regression results. The findings suggest that spending tends to decrease slightly with age, while younger consumers, especially in Footwear and Accessories, exhibit higher average purchases — possibly reflecting trend-oriented behavior. From a storytelling perspective, the scatter plot conveys a subtle yet meaningful narrative: as customers grow older, their purchasing behavior becomes more consistent and value-driven, whereas younger segments show higher variability and fashion sensitivity. These insights support targeted marketing strategies — focusing promotional efforts on trend-based products for younger consumers while emphasizing comfort and practicality for older demographics. Overall, this scatter plot successfully combines analytical precision, interactivity, and visual clarity to deliver actionable behavioral insights for strategic decision-making.

2.4. Revenue Structure by Category and Item

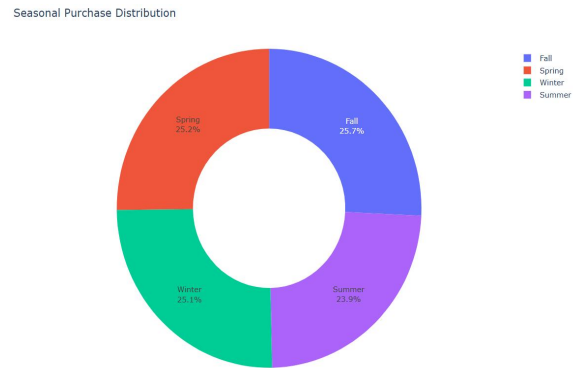
Revenue Structure by Category and Item



- The treemap chart was selected to visualize the company’s Revenue Structure by Category and Item, effectively representing hierarchical relationships (Category → Item) and proportional revenue contributions in a compact layout. Each rectangle corresponds to a specific product item, with its size proportional to revenue and its color indicating the major product category — Clothing, Accessories, Footwear, or Outerwear. This visualization type was chosen because it efficiently combines both macro-level and item-level insights, outperforming pie or bar charts in space utilization and detail density. Within the interactive dashboard, the treemap incorporates several features such as category filters, hover tooltips showing exact revenue and percentage contributions, and drill-down capabilities for deeper exploration (e.g., from “Clothing” to individual items like “Dress” or “Pants”). Additionally, cross-filtering with other visualizations allows users to connect product performance with factors like customer demographics or purchase frequency, making the treemap a dynamic tool for multidimensional analysis.
- From a UX/UI perspective, the chart employs a well-balanced color palette (blue for Clothing, red for Accessories, green for Footwear, and purple for Outerwear), ensuring clear visual differentiation and aesthetic harmony. The layout remains compact and readable even with numerous items, aided by consistent font sizing, spacing, and title hierarchy aligned with dashboard design standards. Regarding data quality and analytical insight, the treemap is based on structured, well-cleaned data that reveals meaningful business patterns: Clothing contributes the largest share of total revenue, followed by Accessories, while Footwear and Outerwear represent smaller portions — potential areas for strategic improvement. These findings help management identify strengths and target underperforming segments for growth initiatives.
- From a storytelling standpoint, the treemap effectively communicates the overarching narrative of the business: “Clothing dominates the company’s sales portfolio, reflecting a fashion-driven customer base, while Accessories also play a strong complementary role. Meanwhile, the lower shares of Footwear and Outerwear indicate opportunities for product diversification and targeted marketing.” Positioned as the opening visualization of the dashboard, the treemap provides an immediate “big picture” overview, setting the foundation for deeper behavioral and demographic analyses. Overall, this treemap successfully integrates

analytical clarity, interactivity, and design aesthetics — transforming raw data into an engaging, insight-driven business story that supports strategic decision-making.

2.5. Seasonal Purchase Distribution



- The Donut Chart was chosen to visualize the Seasonal Purchase Distribution across Spring, Summer, Fall, and Winter, effectively illustrating the proportion of total purchases made during each season. The hollow center enhances focus and readability, allowing for inclusion of total figures or annotations. This visualization type is ideal for showing parts of a whole (100%) while maintaining a clean, modern aesthetic that integrates seamlessly into dashboards. Within the interactive dashboard, the donut chart supports intuitive features such as hover tooltips displaying exact percentages and totals, clickable legends for seasonal filtering, and drill-through navigation to explore category performance within each season. These interactive capabilities transform a simple chart into a dynamic tool for understanding consumer purchase timing.
- From a UX/UI perspective, the chart employs a harmonious, contrasting color palette (blue–Fall, red–Spring, green–Winter, purple–Summer), clear labeling, and balanced circular symmetry, ensuring clarity and visual harmony suitable for executive dashboards. The data analysis reveals that purchasing activity remains stable throughout the year: Fall (25.7%) leads slightly, followed by Spring (25.2%), Winter (25.1%), and Summer (23.9%). This suggests consistent customer engagement and effective year-round marketing performance.
- From a data storytelling standpoint, the donut chart conveys a narrative of stability with mild seasonal variation: “Customer purchasing remains steady across all seasons, with a small peak in Fall and a slight dip in Summer.” This insight provides actionable guidance — reinforcing that the company can leverage the Fall season for new product launches while introducing targeted promotions in Summer to balance performance. Overall, the donut chart successfully combines visual simplicity, interactivity, and analytical storytelling to communicate stable purchasing behavior and support data-driven marketing strategies.

2.6. Distribution of Customer Review Ratings



- The Histogram was used to visualize the distribution of customer review ratings ranging from 2.5 to 5, where each bar represents the number of reviews within a rating interval. This type of visualization is ideal for illustrating the frequency distribution of continuous numerical data and identifying sentiment patterns or skewness. In an interactive Python dashboard, hover tooltips display exact counts and percentages, while filters and dynamic range selection allow users to focus on specific rating intervals or link results with other visuals such as “Product Category by Rating.”

- From a UX/UI perspective, the histogram applies a clear layout with descriptive axes (“Rating” and “Count”), blue-tone color scheme, and minimal visual clutter, ensuring both readability and professionalism. The data analysis indicates that most customer ratings fall between 3 and 4.5, revealing overall positive sentiment and stable satisfaction levels, though the relatively smaller number of 5-star ratings highlights potential areas for refinement.

- The data storytelling conveys that customer satisfaction is strong but can be further enhanced: “Most customers rate between 3 and 4.5, showing consistent satisfaction, while the gap in perfect scores suggests opportunities to elevate experiences.” This insight supports actionable recommendations — such as improving service responsiveness or personalizing customer engagement — to transform “good” reviews into “excellent” ones. Overall, the histogram effectively translates numerical distribution into a clear narrative of customer sentiment and strategic improvement.

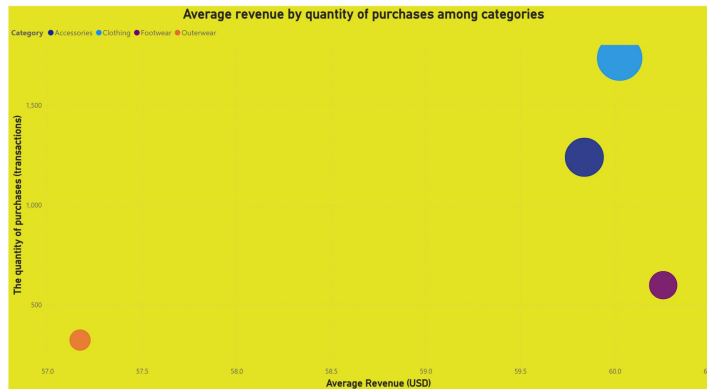
* Results made by Power BI:

- Analysis by the factors 'Category', 'Purchased Item' :



- Techniques used: Horizontal bar chart with the following elements:
 - ◆ Y-axis (Category): Represents the main product categories: Clothing, Accessories, Footwear, and Outerwear.
 - ◆ X-axis (The quantity of purchases): Represents the number of purchases.
 - ◆ Legend for the factor 'Size' with colors: The chart uses different colors to represent the sizes:
 - ✧ Dark Blue: Size L
 - ✧ Light Blue: Size M
 - ✧ Orange: Size S
 - ✧ Purple: Size XL
- Analysis results:
 - ◆ Clothing: This category has the highest quantity of purchases. The majority of orders fall into sizes L (dark blue) and M (light blue), indicating that these are the most popular sizes for clothing. Sizes S and XL have significantly fewer purchases.
 - ◆ Accessories : Accessories also have a large volume of purchases, with size M (light blue) accounting for the largest portion. Size L (dark blue) is the second largest, followed by S (orange) and XL (purple) with similar and smaller quantities.
 - ◆ Footwear: Footwear has a lower purchase quantity compared to clothing and accessories. Size M (light blue) is the best-selling size, followed by L (dark blue). Sizes S (orange) and XL (purple) have relatively small purchase quantities.
 - ◆ Outerwear: This category has the lowest quantity of purchases in the chart. Sizes L (dark blue) and M (light blue) have roughly equivalent and the largest quantities in this group. Sizes S (orange) and XL (purple) have very low purchase quantities.

- Analysis by the factors 'Category', 'Purchased Item' :



- Techniques used: Scatter plot with the following elements:

- ◆ X-axis (horizontal): Average Revenue (USD) — shows how much customers typically spend on average per purchase in each category
- ◆ Y-axis (vertical): The quantity of purchases (transactions) — shows how many transactions or sales occurred in each category.
- ◆ Legend: Category with colors:
 - ✧ Accessories – Dark blue
 - ✧ Clothing – Light blue
 - ✧ Outerwear – Orange
 - ✧ Footwear – Purple

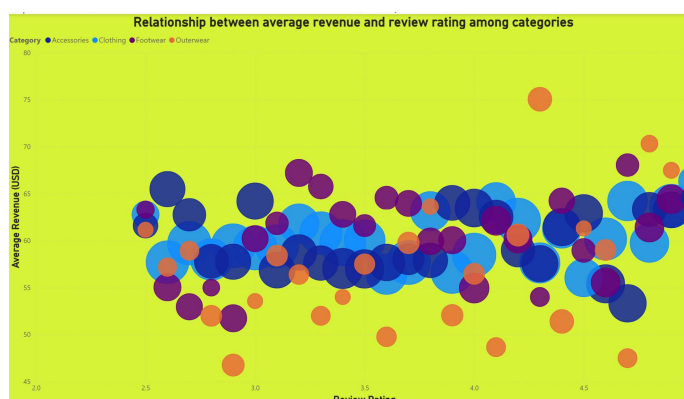
- Analysis results:

- ◆ Clothing (dark blue) stands out as the top-performing category — it has both the highest number of transactions (around 1,700+) and a high average revenue (~\$59).
- ◆ Accessories (light blue) ranks second, showing a moderate number of purchases and slightly lower average revenue (~\$59–60).
- ◆ Footwear (orange) has fewer purchases (around 600) but a higher average revenue per purchase (~\$60), suggesting higher-priced items or less frequent but more valuable sales.
- ◆ Outerwear (purple) has the fewest transactions (below 300) and the lowest average revenue (~\$57), making it the least performing category in both metrics.

=> Insight:

- ◆ Customers buy clothing most frequently, contributing significantly to total sales volume.
- ◆ Footwear might be a premium category, generating higher revenue per sale but with fewer purchases.
- ◆ Outerwear could benefit from promotional strategies or seasonal campaigns to boost both sales and revenue.
- ◆ The chart overall helps identify which categories drive the most sales and which offer higher value per purchase.

- Analysis by the factors 'Category', 'Purchased Item' :



- Techniques used: Scatter plot with the following elements:
 - ◆ X-axis (horizontal): Review Rating — represents how customers rated the products (from low ≈ 2.0 to high ≈ 5.0)
 - ◆ Y-axis (vertical): Average Revenue (USD) — shows the average spending per transaction.
 - ◆ Legend: Category with colors:

- ✧ Accessories – Dark blue
- ✧ Clothing – Light blue
- ✧ Outerwear – Orange
- ✧ Footwear – Purple

- Analysis results:

- ◆ Overall, the average revenue tends to remain between \$55–65 USD across most ratings, meaning there is no strong correlation between review ratings and spending.
- ◆ Higher review ratings (4.0–5.0) are slightly associated with higher revenues, but the difference is not dramatic.
- ◆ Clothing (light blue) and Accessories (dark blue) dominate the chart — indicating these categories have more data points and consistent revenue levels across all ratings.
- ◆ Footwear (orange) shows a few bubbles with very high revenue values ($\sim \$75$ – $\$80$) even at moderate or high ratings, implying premium products that generate higher income despite fewer transactions.
- ◆ Outerwear (purple) appears scattered across different ratings, generally with moderate revenue, suggesting more variability in customer satisfaction and spending.

=> Insights:

- ◆ There is no clear linear relationship between customer satisfaction (review rating) and spending amount.
- ◆ Clothing and accessories are stable categories — customers spend similar amounts regardless of their ratings.
- ◆ Footwear has fewer but more profitable sales, potentially due to higher product prices.
- ◆ Outerwear may need improvements in product consistency or customer experience to achieve better ratings and revenue stability.

- Analysis by the buyer behavior:



- Techniques used: These visualizations analyze shopping behavior by gender (Female vs. Male) using treemap charts filtered with a gender slicer. The two treemaps show:

- ◆ The quantity of purchases (left side)
- ◆ Revenue (USD) generated (right side):
 - ✧ Each block represents a product item grouped under broader categories (Clothing, Accessories, Footwear, Outerwear)
 - ✧ The size of each block indicates its relative contribution to total purchases or revenue

■ Analysis results:

◆ For Female Customers:

- ✧ Clothing is the most purchased and highest-revenue category among female buyers.
- ✧ Top clothing items include *Dress*, *Skirt*, *Pants*, *Hoodie*, and *T-shirt*.
- ✧ Footwear also performs strongly, especially *Sandals* and *Boots*.
- ✧ Accessories (e.g., *Handbags*, *Jewelry*, *Sunglasses*) account for a significant number of transactions but contribute less revenue than clothing.
- ✧ Outerwear (e.g., *Coat*, *Jacket*) has the lowest purchase quantity and revenue share.

=> Insights:

- Female buyers tend to spend more on fashion-related items (Clothing, Footwear, Accessories).
- Clothing items drive both volume and value, indicating they are key to attracting and retaining female customers.
- Accessories may benefit from bundled offers or discount promotions to increase their revenue share.

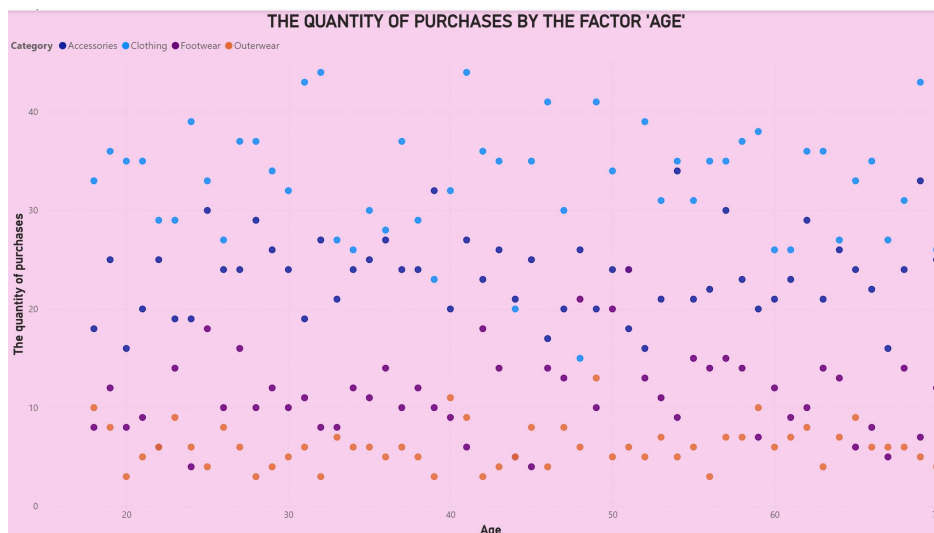
◆ For Male Customers:

- ✧ Similar to females, Clothing remains the top category for both purchase quantity and revenue.
- ✧ Commonly bought items include *Shirts*, *Pants*, *Shorts*, and *Sweaters*.
- ✧ Footwear (especially *Shoes*, *Sneakers*, *Sandals*) shows a strong revenue contribution despite fewer total purchases.
- ✧ Accessories (e.g., *Backpacks*, *Belts*, *Hats*, *Scarves*) are moderately purchased but less profitable.
- ✧ Outerwear (*Coats*, *Jackets*) has the smallest proportion of sales and revenue.

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- Outerwear (*Coats*, *Jackets*) has the smallest proportion of sales and revenue.

● Analysis by the buyer behavior:



■ Techniques used: Scatter plot with the following elements:

- ◆ Axes: The x-axis represents "Age" on a scale from 0 to 70. The y-axis represents "The Quantity of Purchases" on a scale from 0 to 40.
- ◆ Data Points: Each dot represents a data point for a specific purchase quantity at a given age. The points are scattered but form noticeable downward-sloping patterns, indicating trends over age groups.
- ◆ Legend: Categories with colors: The legend labels categories as follows:
 - ✧ Accessories: Purple dots
 - ✧ Clothing: Blue dots
 - ✧ Footwear: Light blue/cyan dots
 - ✧ Outerwear: Orange dots

■ Analysis results:

- ◆ There is a clear negative correlation between age and purchase quantity. Younger buyers (teens to 30s) exhibit higher purchase volumes, often exceeding 30 units, while older buyers (50+) rarely surpass 10-15 units. This suggests diminishing buying activity with age, which could be influenced by factors like lifestyle changes, reduced disposable income, shifting priorities (e.g., from fashion to essentials), or targeted marketing toward youth.
- ◆ Category Insights:
 - ✧ Clothing (blue dots): Concentrated at higher quantities (20-40) in younger age groups (20-40), with a steep decline after age 40. This implies clothing purchases are more frequent or voluminous among young adults, possibly due to trends, social influences, or seasonal buying.
 - ✧ Accessories (purple dots): Similar to clothing but slightly lower overall (peaking around 20-30 units), with points spread across mid-ages (30-50). Accessories might appeal to a broader but still youth-skewed demographic.
 - ✧ Footwear (light blue/cyan dots): Mid-range quantities (10-30), more evenly distributed but still decreasing with age. This category shows less variability, indicating steady but moderate demand across ages, perhaps as a necessity item.
 - ✧ Outerwear (orange dots): Lowest quantities (mostly under 20), dominant in older age groups (40+). This could reflect seasonal or practical needs that don't drive high-volume purchases, with older buyers prioritizing durability over quantity.
- ◆ Potential Implications: The plot highlights opportunities for targeted strategies, such as promoting high-volume items like clothing to younger demographics or focusing on quality outerwear for older groups. Variability in points (e.g., outliers at high ages with moderate quantities) suggests individual differences, but the overarching downward trend points to age as a key factor in buyer behavior. If this data comes from a larger dataset (e.g., e-commerce sales), it could inform inventory planning or personalized marketing.

● Analysis by the buyer behavior:





■ Techniques used:

◆ Gender-Specific Size and Shipping Analysis (Female):

- ✧ Slicer: Allows filtering by gender, with "Female" selected (black checkbox) and "Male" unselected.
- ✧ Stacked Column Chart (Quantity of Purchases Among Sizes): Vertical columns represent purchase quantities for different sizes (M, L, S, XL). Colors correspond to sizes: M (green), L (red), S (purple), XL (yellow). The y-axis scales from 0 to 600. Bars show approximate quantities: M (~550), L (~300), S (~200), XL (~100). Each column is solid and not visibly stacked, suggesting it represents total quantity per size for females.
- ✧ Stacked Bar Chart (Quantity of Purchases Among Shipping Types): Horizontal bars represent purchase quantities for shipping types (Free Shipping, Standard, 2-Day Shipping, Express, Store Pickup, Next Day Air). Colors: Free Shipping (blue), Standard (green), 2-Day Shipping (pink), Express (purple), Store Pickup (dark blue), Next Day Air (orange). The x-axis scales from 0 to 300. Approximate quantities: Free Shipping (~280), Standard (~250), 2-Day Shipping (~200), Express (~220), Store Pickup (~180), Next Day Air (~150). Bars are solid, not stacked.
- ✧ Overall Subtitle: "The quantity of purchases among sizes, shipping types by the gender 'Female'".

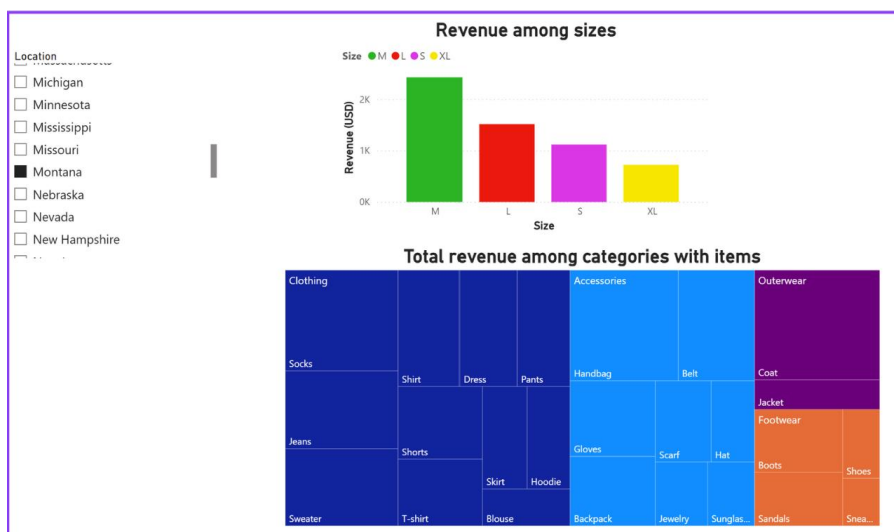
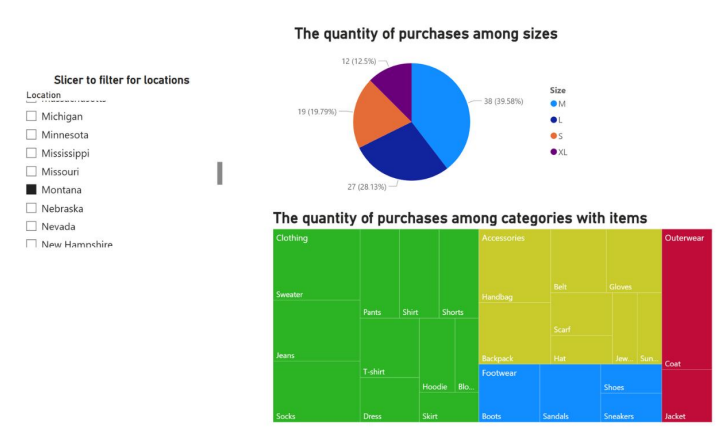
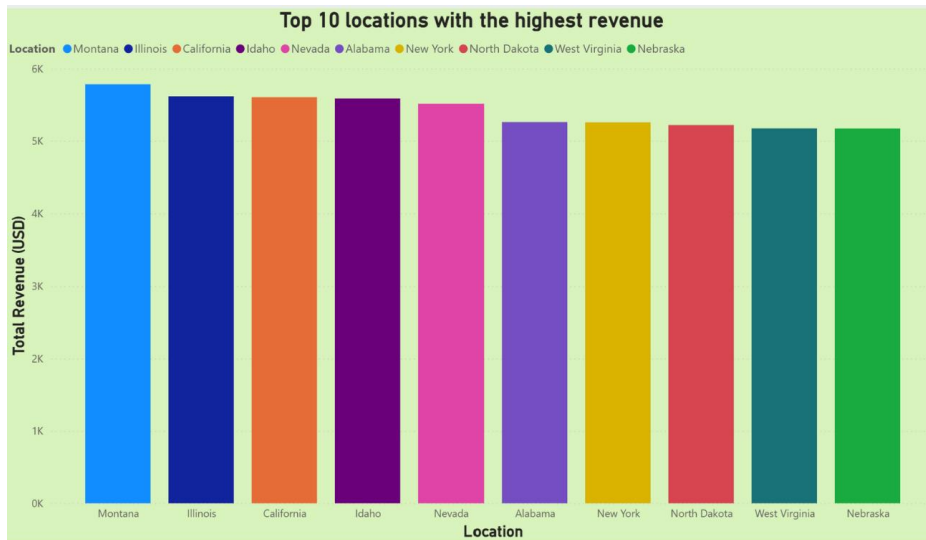
◆ Gender-Specific Size and Shipping Analysis (Male):

- ✧ Slicer: Similar to above, but "Male" selected (black checkbox) and "Female" unselected.
- ✧ Stacked Column Chart (Quantity of Purchases Among Sizes): Similar structure, but y-axis scales to 1,200. Approximate quantities: M (~1,100), L (~600), S (~400), XL (~200). Colors same as female chart.
- ✧ Stacked Bar Chart (Quantity of Purchases Among Shipping Types): X-axis scales to 500. Approximate quantities: Next Day Air (~350), Store Pickup (~400), Express (~400), Standard (~400), Free Shipping (~400), 2-Day Shipping (~400). Colors same as female chart.
- ✧ Overall Subtitle: "The quantity of purchases among sizes, shipping types by the gender 'Male'".

◆ Payment Methods Analysis: Clustered Bar Chart (Quantity of Purchases by Payment Methods Among Genders): Horizontal bars compare quantities across payment methods (Bank Transfer, Venmo, Debit Card, Cash, Credit Card, PayPal) for two genders. Colors: Female (green), Male (blue). The x-axis scales from 0 to 500. For each method, male bars are longer. Approximate quantities:

- ✧ Bank Transfer: Male (~380), Female (~220)
- ✧ Venmo: Male (~420), Female (~220)
- ✧ Debit Card: Male (~420), Female (~180)
- ✧ Cash: Male (~400), Female (~200)
- ✧ Credit Card: Male (~400), Female (~220)
- ✧ PayPal: Male (~450), Female (~250)
- ✧ Legend: Gender differentiated by color.
- ✧ Overall Subtitle: "The quantity of purchases by payment methods among genders".

● Analysis by the feature 'Location':



◆ Stacked Column Chart: Total Revenue by Top 10 Locations

- ✧ Description: This visualization is a column chart (labeled as "stacked column chart," but it appears to function as a standard column chart since each bar is uniformly colored and represents a single location's total revenue without visible stacking). It displays the top 10 locations with the highest total revenue (purchase amounts) in USD. The x-axis lists the locations in descending order of revenue: Montana, Illinois, California, Idaho, Nevada, Alabama, New York, North Dakota, West Virginia, and Nebraska. The y-axis shows Total Revenue [USD], ranging from 0K to 6K. Each bar is colored differently based on the legend (e.g., Montana in cyan, Illinois in indigo, California in orange, etc.), and the bars decrease in height from left to right, indicating a gradual drop in revenue. Montana leads with approximately 5.8K USD, while Nebraska is at the bottom with around 4.2K USD.
- ✧ Explanation and Insights: This chart highlights geographic performance in terms of revenue generation. Montana emerges as the top-performing location, contributing the highest purchase amounts, possibly due to higher customer spending, larger order values, or a denser customer base in that area. The close clustering of revenues (all between ~4K and 6K USD) suggests relatively even distribution among these top locations, with no single outlier dominating excessively. Differences could stem from factors like regional economic conditions, marketing effectiveness, or product preferences. For instance, states like Montana and Illinois might have stronger demand or higher average transaction values compared to others. This could inform business decisions, such as targeting marketing campaigns or inventory allocation to these high-revenue areas to maximize profits.

◆ Bar Chart: Total Quantity of Purchases by Top 10 Locations:

- ✧ Description: This is a horizontal bar chart showing the top 10 locations with the most purchases (measured by Customer_ID, likely indicating the number of unique customers or transactions). The y-axis lists locations in descending order: Montana, California, Idaho, Illinois, Alabama, Minnesota, Nebraska, Nevada, New York, Delaware, and Maryland. The x-axis represents "The quantity of purchases," ranging from 0 to 100. Bars extend rightward, with Montana having the longest bar at around 98 purchases, followed closely by California at ~95, and Maryland at the bottom with ~82. Each bar is colored per the legend (e.g., Montana in blue, California in orange, Idaho in purple), emphasizing volume over value.
- ✧ Explanation and Insights: This chart focuses on transaction volume rather than monetary value, revealing where the highest number of purchases occur. Montana again tops the list, indicating it's not only a revenue leader but also a high-volume market, suggesting efficient customer acquisition or repeat business. Notably, the order differs slightly from the revenue chart (e.g., California ranks higher here than in revenue, while Illinois drops a few spots), which implies variations in average purchase size—locations like Illinois might have fewer but higher-value transactions, whereas California has more frequent but possibly lower-value ones. This discrepancy could point to opportunities for upselling in high-volume areas or pricing adjustments in high-revenue but lower-volume spots. Overall, these top locations account for the bulk of activity, potentially representing key markets for expansion or loyalty programs.

◆ Pie Chart and Treemap: Quantity of Purchases in Montana (by Size and Category):

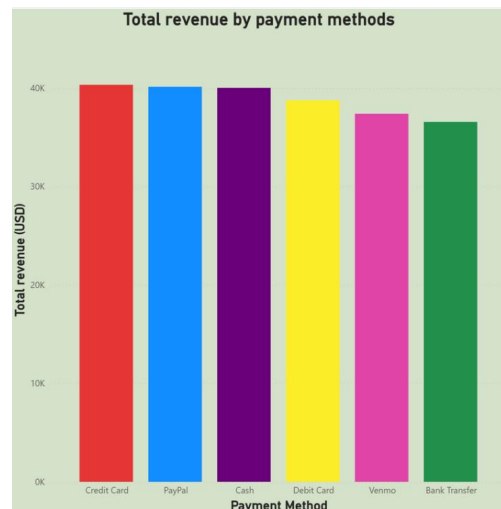
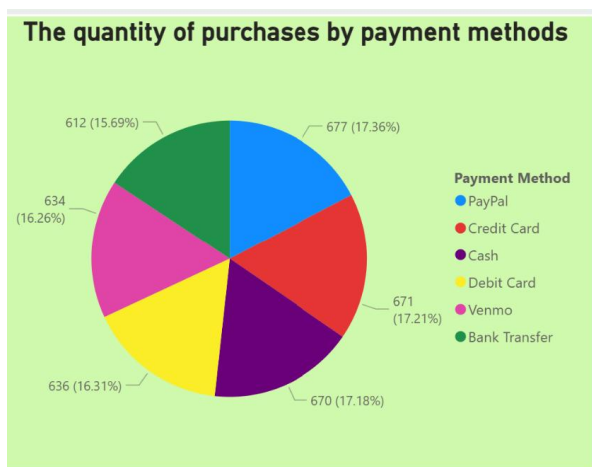
- ✧ Description: This dashboard combines a slicer (checkbox list for filtering locations, with Montana selected), a pie chart, and a treemap, all focused on purchase quantities in Montana.
- ✓ Pie Chart: Titled "The quantity of purchases among sizes," it shows distribution by clothing sizes: M (38.9%, blue), L (27.8%, purple), S (19.7%, orange), XL (12.5%, dark blue). The pie is segmented proportionally, with M occupying the largest slice.
- ✓ Treemap: Titled "The quantity of purchases among categories with items," it uses rectangular blocks sized by quantity. Categories include Clothing (green, largest group with sub-items like Sweater, Pants, Shirt, etc.), Accessories (yellow, with Handbag, Belt, etc.), Outerwear (red, with Coat, Jacket), and Footwear (blue, with Boots, Shoes, etc.). Clothing dominates the space, followed by Accessories.

- ✧ **Explanation and Insights:** Filtered to Montana (the top location from previous charts), this provides a drill-down into product-level details by quantity. The pie chart reveals a preference for medium (M) sizes, which make up nearly 40% of purchases—possibly reflecting the average body size in the region or targeted inventory stocking. Smaller sizes (S and XL) are less popular, suggesting potential overstock risks or opportunities to promote them. The treemap illustrates category dominance: Clothing leads in volume, indicating it's the core product line driving purchases in Montana, with everyday items like Sweaters and Pants being top sellers. Accessories and Footwear follow, while Outerwear is smaller, perhaps due to seasonal factors (e.g., less demand in warmer months). This insight could guide inventory management, such as stocking more M-sized clothing items, or marketing bundles combining popular categories to boost sales in this high-performing location.

◆ **Stacked Column Chart and Treemap: Total Revenue in Montana (by Size and Category):**

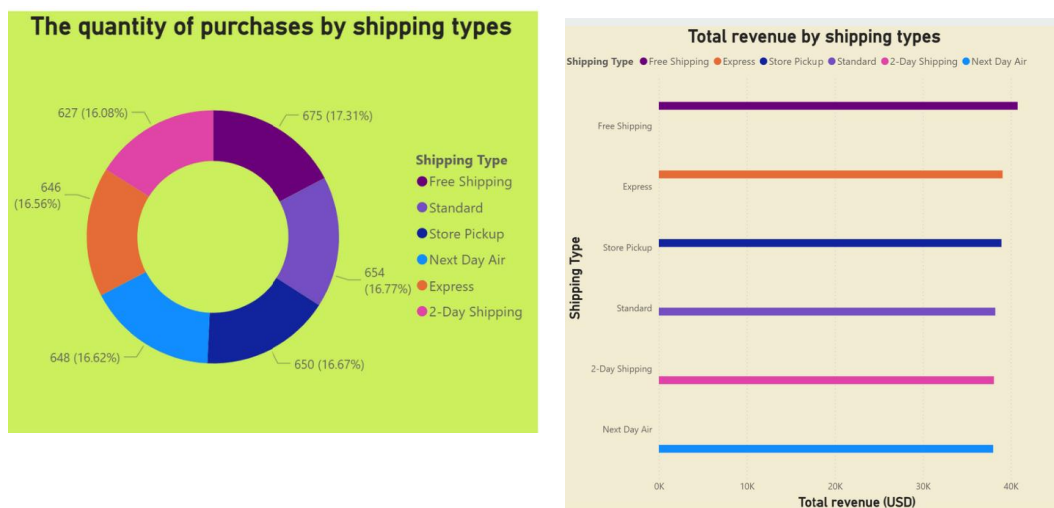
- ✧ **Description:** Similar to the third visualization, this includes a location slicer (Montana selected), a stacked column chart, and a treemap, but focused on revenue (USD) rather than quantity.
 - ✓ **Stacked Column Chart:** Titled "Revenue among sizes," it shows columns for sizes M (green, ~250 USD), L (red, ~200 USD), S (purple, ~150 USD), and XL (yellow, ~100 USD). Each column is stacked by color-coded sub-elements (though minimal stacking is visible, it might represent breakdowns within sizes). M has the tallest stack.
 - ✓ **Treemap:** Titled "Total revenue among categories with items," blocks are sized by revenue. Categories: Clothing (dark blue, largest, with sub-items like Socks, Shirt, etc.), Accessories (cyan, with Handbag, Gloves, etc.), Footwear (orange, with Boots, Shoes, etc.), and Outerwear (purple, with Coat, Jacket). Clothing again dominates.
- ✧ **Explanation and Insights:** This revenue-focused view on Montana complements the quantity analysis, showing how volume translates to dollars. The stacked column chart indicates M sizes generate the most revenue (~250 USD), aligning with their high quantity share, but with possible higher pricing or margins contributing. Smaller sizes like XL yield less, reinforcing the need for size-specific strategies. The treemap mirrors the quantity one but emphasizes monetary impact: Clothing drives the most revenue, suggesting it's not only popular in volume but also profitable (e.g., higher-priced items like Sweaters or Jeans). Accessories and Footwear contribute moderately, while Outerwear lags, potentially due to lower unit prices or seasonal sales dips. Comparing to the quantity charts, revenue patterns are similar but amplified in high-margin categories—indicating that while Montana has high purchase counts, profitability is concentrated in core clothing items. Businesses could use this to optimize pricing, promotions, or supplier focus on top-revenue drivers in this location.

● **Analysis by the feature 'Payment Method':**



- **Description:** This visualization consists of two parts: a pie chart on the left showing the distribution of purchase quantities by payment method, and a column chart (labeled as "stacked column chart" but appearing as standard columns without visible stacking) on the right displaying total revenue in USD by payment method. The pie chart includes six payment methods with their quantities and percentages: Credit Card (red, 677 purchases, 17.36%), Bank Transfer (green, 670 purchases, 17.18%), Debit Card (yellow, 671 purchases, 17.21%), Cash (purple, 636 purchases, 16.31%), Venmo (pink, 634 purchases, 16.26%), and PayPal (blue, 612 purchases, 15.69%). The slices are roughly equal, indicating a balanced distribution. The column chart lists methods on the x-axis in descending revenue order: Credit Card (~38K USD, red), PayPal (~35K USD, blue), Cash (~32K USD, purple), Debit Card (~30K USD, yellow), Venmo (~28K USD, pink), and Bank Transfer (~25K USD, green). The y-axis ranges from 0K to 40K USD, with bars decreasing slightly from left to right.
- **Explanation and Insights:** The pie chart reveals a relatively even spread of transaction volumes across payment methods, with each accounting for about 16-17% of purchases, suggesting diverse customer preferences without a dominant option. This balance could reflect effective integration of multiple payment gateways, accommodating various user demographics (e.g., tech-savvy users favoring PayPal or Venmo, while others prefer traditional cards). However, the revenue chart shows more variation: Credit Card generates the highest revenue despite similar transaction counts, implying higher average order values—possibly due to customers using credit for larger purchases or associated promotions. Conversely, Bank Transfer yields the least revenue, which might indicate lower-value transactions or fees deterring big spends. This discrepancy highlights opportunities for optimization, such as incentivizing high-revenue methods (e.g., rewards for credit card use) or analyzing why certain methods underperform in value despite volume parity, potentially improving overall profitability.

● **Analysis by the feature ‘Shipping Type’:**



- **Description:** This setup includes a pie chart on the left for purchase quantities by shipping type and a horizontal bar chart (labeled as "stacked bar chart" but showing uniform bars without clear stacking) on the right for total revenue in USD. The pie chart covers six shipping types with quantities and percentages: Store Pickup (dark blue, 654 purchases, 16.77%), Next Day Air (light blue, 648 purchases, 16.62%), 2-Day Shipping (pink, 627 purchases, 16.08%), Standard (blue, 650 purchases, 16.67%), Free Shipping (purple, 675 purchases, 17.31%), and Express (orange, 646 purchases, 16.56%). Slices are very similar in size, showing even distribution. The bar chart has shipping types on the y-axis (Free Shipping at top, Next Day Air at bottom) and revenue on the x-axis from 0K to 40K USD. All bars are nearly identical in length, around 30K USD each, with slight variations (e.g., Free Shipping slightly higher, Next Day Air slightly lower), colored per the legend.

- Explanation and Insights: The pie chart demonstrates uniform adoption of shipping options, with each type comprising roughly 16-17% of purchases, indicating no strong bias toward faster or cheaper methods—customers seem equally open to convenience (e.g., Store Pickup), speed (e.g., Next Day Air), or cost savings (e.g., Free Shipping). This could stem from competitive pricing, seasonal factors, or broad appeal across regions. The revenue bar chart, however, shows remarkably consistent earnings across types, with minimal differences despite volume parity, suggesting average order values are similar regardless of shipping choice. This implies shipping doesn't significantly influence basket size; for instance, premium options like Express don't correlate with higher spends, possibly because costs are passed to customers or offset by promotions. Businesses might leverage this by promoting underutilized fast-shipping for urgency-driven sales or enhancing free shipping thresholds to boost volumes without revenue dilution, ultimately refining logistics and customer satisfaction strategies.

IV. CONCLUSIONS

1. Analysis Results

The analysis of the "shopping_behavior.csv" dataset reveals key patterns in online shopping behavior among US customers in 2025, derived from a combination of demographic, transactional, and operational factors. These insights were uncovered through data visualizations created using Python and Power BI, focusing on variables such as product categories, customer demographics (age and gender), seasonal trends, regional performance, and preferences in payment and shipping methods.

- Top-performing categories: Accessories and Clothing dominate total revenue, showing strong consumer demand: From the treemap and bar chart visualizations, Clothing emerges as the leading category in both purchase quantity and total revenue, accounting for the largest share of sales across the dataset. This is evident in hierarchical breakdowns where sub-items like Sweaters, Pants, Shirts, Hoodies, and T-Shirts within Clothing contribute significantly to overall earnings. Accessories follow closely as the second-highest performer, with items such as Handbags, Belts, Gloves, Scarves, Hats, Backpacks, Jewelry, and Sunglasses driving substantial demand. Footwear and Outerwear lag behind, representing smaller proportions of revenue—Footwear with sub-items like Boots, Shoes, Sandals, and Sneakers, and Outerwear with Coats and Jackets. This dominance of Clothing and Accessories suggests a robust consumer preference for everyday essentials and complementary items, potentially influenced by fashion trends, affordability, and versatility. The even distribution in some visualizations highlights that while these categories lead in volume, their average purchase values remain competitive, indicating broad appeal rather than reliance on high-margin luxury sales.
- Customer demographics: Older age groups tend to spend higher amounts per purchase, while younger customers shop more frequently: Scatter plots examining the relationship between age and purchase amount, segmented by category, show a slight downward trend in spending as age increases, but with notable nuances. Younger customers (typically under 30) exhibit higher frequency of purchases, as seen in denser clustering of data points in the 20-40 purchase quantity range for categories like Footwear and Accessories. This suggests that millennials and Gen Z shoppers engage more often, possibly due to impulse buying, social media influences, or lower barriers to entry via mobile shopping. In contrast, older age groups (over 50) demonstrate higher average spend per transaction, with sparser but elevated purchase amounts, particularly in Clothing and Outerwear. This could stem from greater disposable income, preference for quality over quantity, or life-stage needs (e.g., professional attire or durable goods). Overall, the data indicates a generational shift: youth-driven volume versus maturity-driven value, with purchase quantities generally decreasing across age groups in a downward-sloping pattern.
- Gender-based behavior: Female customers contribute more to the total number of transactions, while male customers make higher-value purchases: Treemap visualizations filtered by gender reveal distinct shopping patterns. For female customers, Clothing dominates both quantity and revenue, with high transaction volumes in items like Dresses, Skirts, Pants, Hoodies, and T-Shirts, followed by Footwear (e.g., Boots and Shoes). This results in females accounting for a larger share of overall transactions, reflecting broader engagement across categories and potentially more frequent, varied shopping sessions. Male customers, however, show similar category preferences

(Clothing top, with Shirts, Pants, Shorts, and Sweaters leading), but with a focus on higher-value purchases—evidenced by elevated average revenue per transaction despite fewer total buys. Accessories and Outerwear play smaller roles for both genders, but males exhibit slightly stronger revenue in practical items like Jackets. These differences may arise from societal norms, marketing targeting, or product assortments, with females driving volume through exploratory shopping and males emphasizing efficiency and larger spends.

- **Seasonal patterns:** Purchase activities peak during holiday seasons, reflecting strong seasonal demand: Donut charts and pie charts illustrating purchase distribution by season highlight clear peaks in Fall and Winter, which align with holiday periods (e.g., back-to-school, Thanksgiving, Christmas). These seasons capture the highest proportions of total purchases and revenue, with Fall recommended for new product launches due to elevated activity. Summer, conversely, shows the lowest performance, possibly due to outdoor activities reducing online engagement or seasonal irrelevance of certain products like Outerwear. Spring maintains moderate levels, serving as a transitional period. This cyclical pattern underscores the impact of external factors like weather, events, and promotions, with interactive filters in Power BI dashboards allowing drill-downs to confirm category-specific spikes (e.g., Accessories in Winter for gifting).
- **Regional trends:** The states of Montana generates the highest revenue, suggesting potential for targeted marketing campaigns: Stacked column and bar charts of top 10 locations by revenue and purchase quantity position Montana as the leader, with approximately 5.8K USD in revenue and 98 transactions, followed closely by Illinois, California, Idaho, and others like Nevada and Alabama. This dominance in both metrics indicates strong market penetration, possibly due to demographic factors, economic conditions, or effective local advertising. Drill-downs into Montana via pie charts and treemaps further show preferences for Medium (M) sizes (38.9% of purchases) and Clothing as the core category. Other high performers like California exhibit high volume but slightly lower per-purchase value, while emerging states (e.g., Nebraska, West Virginia) show potential for growth. These geographic insights reveal uneven distribution, with no single factor explaining variations, but opportunities for region-specific strategies.
- **Payment and shipping preferences:** Credit cards and PayPal delivery remain the most popular choices among customers: Pie charts for purchase quantity and column charts for revenue by payment method show a balanced distribution, with each method (Credit Card, Bank Transfer, Debit Card, Cash, Venmo, PayPal) comprising 15-17% of transactions. However, Credit Cards lead in revenue (around 38K USD), suggesting higher average order values, while PayPal follows closely at 35K USD, favored for its convenience and security in online transactions. Shipping preferences, via pie and bar charts, are similarly uniform (16-17% per type: Store Pickup, Next Day Air, 2-Day Shipping, Standard, Free Shipping, Express), with revenue consistently around 30K USD each, indicating no strong correlation between speed/cost and spend. This even adoption reflects diverse customer needs, from urgency (Express) to cost-savings (Free Shipping), without significant impact on basket size.

2. Business Recommendations

Based on the analysis results, the following recommendations aim to capitalize on identified trends, optimize operations, and drive growth for the retail store. These strategies focus on targeted marketing, personalization, inventory management, and payment incentives to enhance customer engagement and profitability.

- **Enhance marketing in high-revenue regions (California, Montana) and expand product reach to other emerging states:** Prioritize digital campaigns in top performers like Montana and California, leveraging location-based ads on social media and email to promote Clothing and Accessories. For Montana specifically, emphasize M-sized items and bundles to align with local preferences. Simultaneously, invest in market expansion to states like Idaho, Nevada, and Alabama through partnerships with local influencers or pop-up events, aiming to replicate success by analyzing demographic overlaps (e.g., younger shoppers in emerging areas).
- **Personalize promotions by demographic:** offer loyalty rewards for older consumers and flash deals to attract younger shoppers: Tailor offers based on age and gender insights—implement tiered loyalty programs for older customers (e.g., exclusive discounts on higher-value Clothing items) to encourage repeat high-spend purchases. For younger demographics, deploy flash sales and social media challenges focused on Accessories and Footwear to boost frequency. Gender-specific personalization could include curated recommendations: value bundles for males (e.g., Shirt-Pants combos) and variety packs for females (e.g., Dress-Accessory sets), tracked via customer profiles for A/B testing.

- Optimize inventory and logistics during holiday seasons to meet peak demand efficiently: Ramp up stock for Clothing and Accessories in Fall and Winter, using predictive analytics from seasonal patterns to avoid stockouts. Enhance logistics by promoting Free Shipping or Store Pickup during peaks to reduce costs, while offering Express options with minimal surcharges. Integrate real-time dashboards to monitor demand spikes, ensuring efficient allocation across regions like Montana.
- Encourage credit card transactions through cashback or installment options to increase conversion rates: Capitalize on Credit Card's high-revenue potential by partnering with providers for cashback (e.g., 5% on purchases over \$100) or zero-interest installments, targeting categories like Clothing. Promote PayPal similarly for tech-savvy users, while analyzing underperformers like Bank Transfer to introduce incentives, ultimately aiming to shift more transactions to high-value methods and reduce cart abandonment.
- Focus on top-selling categories by launching exclusive products or bundles in Accessories and Clothing: Develop exclusive lines or bundles (e.g., T-Shirt + Handbag sets) to amplify demand in these dominant categories, incorporating size preferences (e.g., more M options). Use insights from gender and age breakdowns to customize—youth-oriented trendy Accessories or durable Clothing for older buyers. Monitor performance via updated visualizations to iterate, potentially expanding to hybrid bundles with Footwear for cross-category growth.